SOUMENDRA GANGULY

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SKILLS

C, Python, Linux	14+ yrs
POSIX shell, Bash	12+ yrs
***NIX sys/net admin**	12+ yrs
SageMath	10+ yrs
SymPy, NumPy	10+ yrs
LaTeX	14+ yrs
GNU Emacs	11+ yrs
HTML, JavaScript	10+ yrs
CSS, Bootstrap	10+ yrs
Git, Docker	8+ yrs
Apache, NGINX	8+ yrs
Django, Flask	8+ yrs
Hugo, Jinja	3+ yrs
Lark parser	1.5 yrs
OpenAI API	2+ yrs
PyTorch	1+ mth
Stable Diffusion	1+ mth

PROJECTS

- github.com/8vasu/sg-diffusion
  PyTorch, Stable Diffusion, Docker, CUDA
- github.com/8vasu/2windows.el
  GNU Emacs, ace-window

EDUCATION

2019–2023, Texas A&M University
PhD, Mathematics

2017–2019, Clemson University
MS, Mathematical Sciences

2014–2016, Chennai Math. Inst.
MSc, Mathematics

2011–2014, Chennai Math. Inst.
BSc, Math and Computer Sci.

LANGUAGES

English, Hindi, Bangla

Danish DU3, Module 3

PROFILE

I am an experienced mathematician and computer scientist with a strong background in number theory and a vast range of programming and system administration skills developed over a decade and a half. During my mathematical research, I have focused on studying analytic behavior of automorphic L-functions.

Independently, I have implemented significant features in several major open-source projects by adding new functions, implementing unit tests, and fixing existing bugs. I have extensive experience in web development. I am also proficient in system administration on Linux and other *nix platforms.

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher in Mathematics at Aarhus University — Sep 2023–NOW
Research

- **Problem solving, analytical reasoning, creativity, initiative:** Employed rigorous proof techniques to analyze and solve complex, abstract problems in the theory of automorphic L-functions, contributing to the understanding of their underlying structures.
- **Leadership, project management:** I am the principal organizer and driving force behind a 2-week international event: The Aarhus Automorphic Forms Conference and Summer School, August 4–15, Aarhus University. This involves managing a team of 9 organizers, coordinating 21 invited speakers, 12 contributed talk speakers, overseeing the logistics of the event including accommodation, transportation, dining, visa applications, and managing budget, time, and funding applications for 100+ participants from around the world.
- **Data analysis and communication:** Interpretation and presentation of data and research results to small and large groups. My academic CV is available on my website linked in the header.
- **Research:** Performed literature review and analysis of existing research papers. Wrote a convenient command-line arXiv preprint batch downloader: github.com/8vasu/paper.py.

Independent programmer — Dec 2023–NOW
Collaboration with SymPy

- **Feature development:** Added LaTeX matrix and complex number parsing capability to the official **SymPy** project in a significant pull request.
- **Package development:** Developed github.com/8vasu/compuTeX, a GNU Emacs package enabling users to input complex mathematical expressions in LaTeX format and output the evaluated expression back in LaTeX with a single keystroke, powered by SymPy.
- **Tools:** Python, SymPy, the Lark parser with EBNF grammar, regular expressions, LaTeX , GNU Emacs, Emacs Lisp, Git.

Independent programmer — Sep 2020–NOW
Collaboration with CPython, FreeBSD, NetBSD, and util-linux

- **Feature development, long-term sustained contribution:**
  - Added 7 new functions to the *os*, *termios*, *tty*, and *pty* modules of the official **CPython** project in an ongoing GitHub issue containing 20 pull requests over 5+ years, significantly enhancing the Python standard library's functionality.
  - Added 2 functions to the standard C library of the **FreeBSD** operating system and wrote the manual page for them.
  - Improved the standard *script(1)* and *scriptreplay(1)* programs on FreeBSD, NetBSD, and util-linux as demonstration of these functionalities.
- **Bug fixing:** Addressed and resolved pseudoterminal-related issues within the CPython project.
- **Software testing:** Created comprehensive **unit tests** to validate the reliability and correctness of newly introduced features.
- **Tools:** C, Python, POSIX/BSD/GNU pseudoterminal APIs, unit testing, Git, Docker, GNU Autotools, *nroff(1)*, *script(1)*, *scriptreplay(1)*.